

**Department of Computer Science
Faculty of Informatics
Addis Ababa University**

Entrance Examination for the M. Sc. Program in Computer Science

Date: June 30, 2006
Time allowed: 3 hours

Name: _____

Instructions

1. Read the instructions and the questions carefully.
2. Write your name **only** on this page. Writing your name on the other pages will automatically disqualify you from this exam.
3. Answer all questions eligibly on the spaces provided on the question papers. Please make sure that sentences are short, grammatically and logically correct. You can use the back side of each paper when space is not sufficient. In such a case, please indicate so.
4. This examination booklet consists of **11** pages (excluding this cover page). Make sure that the question papers are numbered consecutively and that no question papers are missing or replicated.
5. Cheating is strictly prohibited. Any person found cheating will be automatically disqualified from the exam.
6. Use only the rough papers provided. Using other unauthorized papers will be considered as cheating.
7. Switch **off** your mobile phones. You are not allowed to use a mobile for **any** purpose.
8. You are not allowed to use calculators or any other electronic device.

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	Part	Part I (35)	Part II (25)	Part III (20)	Part IV (20)	Total (100)
	Marks					

Part I: Programming, Data Structures, and Software Engineering (35 points)

1.1 Discuss briefly software process, project and product. (6 points)

1.2 a. Define coupling and cohesion. (2 points)

b. How are coupling and cohesion used in modularization criteria? (2 Points)

- 1.3 Write a pseudo code for a function ***file_copy*** that copies the content of one file to a new file. The function takes one argument, *original_file*, which is the path to the file to be copied and returns the name of the new file. If an error occurred, NULL should be returned. The function will have the following signature:

function **file_copy** (*original_file* : string) : *copy_file* : string

(7 points)

To write the function, use the following built-in pseudo functions for file manipulation:

fopen(*file_path* : string, *mode* : char) : integer – opens the file whose path is passed to it as an argument and returns a file handle and sets the read pointer at the first byte of the file. On error, it returns -1. *mode* indicates the mode in which the file is opened and it can take one of the following values: **r**, **w** which stand for read and write, respectively.

fclose(*file_handle* : integer) – closes the file whose file handle is passed to it as an argument. On error, it returns -1.

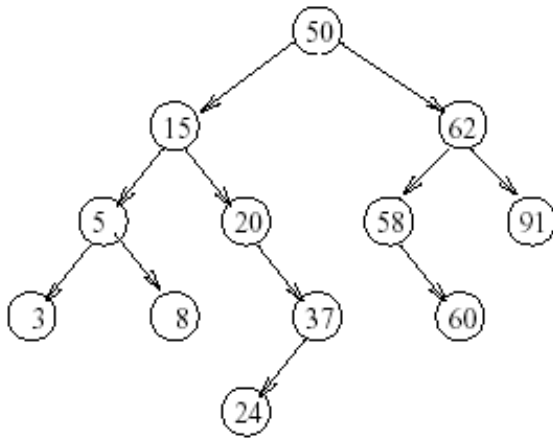
fread(*file_handle* : integer) : ByteStream – reads up to 1 KB of stream of bytes starting from the current position of the read pointer. On error, it returns -1.

fwrite(*file_handle*: integer, data: ByteStream) : integer – writes up to 1 KB of stream of bytes to the file whose handle is passed as an argument at the current end of file. On error, it returns -1.

eof(*file_handle* : integer) : boolean – returns TRUE if end of file is encountered, otherwise returns FALSE.

The answer for question 1.3 may continue on this page.

- 1.4 Consider the following **binary search tree**: If the root node, 50, is deleted, which node will become the new root? (4 points)



Answer: _____

- 1.5. Given the sorting algorithm represented in the following pseudo code, attempt the questions that follow. Assume **A** has **n** elements, where the first index is **1** and the last index is **n**.

```
1: for i ← n-1 downto 1
2:   begin
3:     sorted ← TRUE
4:     for j ← 1 to i
5:       begin
6:         if A[j] < A[j + 1] then
7:           SWAP(A[j], A[j + 1])
8:           sorted ← FALSE
9:         end
10:    if sorted then return
11:  end
```

- a. What is the best case for this algorithm? What is the best case computing time in “big *O*” notation? (3 points)
- b. What is the worst case and corresponding computing time for this algorithm? (3 points)
- c. What is the name of the sorting algorithm represented with the above pseudo code? (3 points)

1.6 Consider a linked list with data values sorted in descending order. The following is given as a sample.

First →

20	→	15	→	9	→	5	→	3	→	
----	---	----	---	---	---	---	---	---	---	--

Write an algorithm that inserts an element in the appropriate position. Your algorithm must handle all the special cases. Make your writing clear; otherwise marks will be deducted. (5 points)

Part II: Computer Architecture and Operating Systems (25 points)

- 2.1 For interactive operating systems, write the advantages of preemptive scheduling algorithms over the non-preemptive scheduling algorithms. (3 points)
- 2.2 Explain why a cache memory, in addition to RAM (or primary memory), is needed in computer systems. (2 points)
- 2.3 In process swapping, write briefly how important **process relocation** is. (3 1/2 points)
- 2.4 Give two possible methods that are used to reduce disk access time. (2 points)
- 2.5 Consider the Intel 8086/8088 microprocessor architecture to answer the following questions: (3 points)
- What is the name of the unit responsible for getting the instructions from memory and loading it in the Instruction Queue for execution? _____
 - What type of addressing mode represents the instruction **MOV CL, [BX][DI]+8** ?

 - If **CS = 24F6** and **IP = 634A**, the physical address is _____.

2.6 For each of the following questions, circle the letter of the correct answer. (1 point each)

- i. The amount of time required to read a block of data from a disk into memory is composed of seek time, rotational latency, and transfer time. Rotational latency refers to
 - a. the time it takes for the platter to make a full rotation.
 - b. the time it takes for the read-write head to move into position over the appropriate track.
 - c. the time it takes for the platter to rotate the correct sector under the head.
 - d. none of the above.
- ii. Interrupts can be generated in response to
 - a. detected program errors such as arithmetic overflow or division by zero.
 - b. detected hardware faults.
 - c. Input/Output activities.
 - d. internal timers.
 - e. all of the above.
 - f. all except a.

2.7 If a magnetic disk has 100 cylinders, each containing 10 tracks of 10 sectors, and each sector can contain 128 bytes, what is the maximum capacity of the disk in bytes? _____ (2 point)

2.8 The computer system bus is made up of three independent bus types. What are they? (3 points)

2.9 A computer's memory is composed of 8K words of 32 bits each. How many bits are required for memory address if the smallest addressable memory unit is a word? _____ (2 point)

2.10 Whose responsibilities are the following tasks to the instructions of the computer system? ((**a**)Execution Unit or (**b**)Bus Interface Unit); Write **a** or **b** on the spaces provided: (1/2 point each)

Fetching: _____
Pre-fetching: _____
Decoding: _____
Execution: _____
Memory Address Generation: _____

Part III: Database Management Systems (20 points)

3.1 Based on the description given below, answer the questions that follow.

The Cheapest Car Company sells new and used cars from different makers such as Ford, Volvo and Toyota and of different models such as Camry, Falcon, Corolla, Carina. *custNo*, *serialNo* and *staffNo* are the sole keys of **Customer**, **Car** and **Staff**, respectively. Customers make orders to buy cars; a customer can make several orders. Each order has a unique order number (*orderNo*) and only one car is ordered in one order. An order is processed by one staff member.

Customer(*custNo*, *custName*, *street*, *city*, *postcode*)

Car(*serialNo*, *model*, *year*, *maker*, *price*)

Staff(*staffNo*, *staffName*, *position*)

Order(*orderNo*, *staffNo*, *date*, *custNo*, *serialNo*, *amount_paid*)

Attribute *amount_paid* of relation **Order** is the amount that a customer pays for an order; a Customer may not pay in full for the order. The other attributes are self-explanatory.

- a. Discuss the likely Functional Dependencies in **Order**. (2 points)

- b. Give the foreign keys in **Order** and the relevant referential integrity constraints. (2 points)

- c. Write an SQL query for each of the following:
 - i. In how many cities does the company have customers? (2 points)

 - ii. List the details of customers whose name contains the string “Wolde”. (2 points)

iii. Cars with price greater than Birr 280,000 are luxury cars. List the customer number, name and address of customers who have ordered more than one luxury car. (3 points)

iv. List the name and position of staff members who have only sold cars to customers living in the cities “Addis Ababa” or “Mekelle”. (3 points)

3.2 When is the concept of a weak entity used in database modeling? (2 points)

3.3 Discuss the various reasons that lead to the occurrence of null values in relations? (2 points)

3.4 What is a data dictionary? Use an example to explain the type of information that is kept in the data dictionary. (2 points)

Part IV: Computer Networks and Data Communications (20 points)

4.1 Compare and contrast **virtual circuit switching** and **datagram switching** based on factors such as addressing (addressing information in every packet), routing and effects of node failures. (3 points)

4.2 For each of the following questions, circle the letter of the correct answer. (1 point each)

- i. Which of the following is a deterministic medium access control protocol?
 - a. ALOHA/Slotted ALOHA
 - b. Token passing
 - c. CSMA
 - d. CSMA/CD
- ii. In a data communication network, a sender has a sliding window of size 15 and the first 15 packets have been sent. The first ACK received by the sender was ACK 5, which frame(s) has the receiver received?
 - a. Frame 5
 - b. Frame 4
 - c. Frame 0 to 2
 - d. Frame 0 to 4
- iii. A transport-layer address is called
 - a. MAC address
 - b. Port number
 - c. IP address
 - d. None of the above

4.3 Consider the TCP/IP model. Write the name of the layer that is responsible for each of the following tasks. (1 point each)

- a. Process-to-process data delivery. _____
- b. Provides user services such as e-mail and file transfer. _____
- c. Mechanical, electrical, and functional interface. _____
- d. Responsible for delivery between adjacent nodes. _____
- e. Responsible for choosing routes for packets. _____

4.4 What is the difference between **error detection** and **error correction**? Which method has a higher overhead? (3 points)

4.5 Consider the two transport layer protocols, TCP and UDP. Which protocol is the most appropriate for each of the following two applications? Justify your answers. (4 points)

a. Money transfer

b. Multimedia

4.6 Give one major reason why standardization is required in computer networking. (2 points)